

STATEMENT OF WORK

Project Title: Technical Assistance for Reduction Gear Troubleshooting and High Speed Shaft Aft Seal Replacement on CGC WALNUT (WLB-205)

Purchase Request: 52000PR260023409

Date: 26FEB2026

1.0 Background

The Coast Guard Cutter WALNUT (WLB-205) requires maintenance on the main reduction gear, Model NO: 96DHCMG-CP. This work involves troubleshooting the system and replacing the High Speed (HS) Shaft Aft Seals. The procedures for this maintenance are outlined in the Philadelphia Gear Technical Manual TPN: 3523, SWBS: 241-A, applicable to hulls 201-205. The Coast Guard will perform the labor and provide all necessary materials and tools.

2.0 Objective

The objective of this Statement of Work is to obtain a qualified technician from Philadelphia Gear to provide expert technical assistance to troubleshoot a significant operational leak in the main reduction gear and to provide oversight for the replacement of the High Speed Shaft Aft Seals onboard CGC WALNUT.

3.0 Scope of Work

The contractor technician shall provide expert technical guidance and assistance to Coast Guard personnel to troubleshoot and correct a significant leak from the reduction gear observed during operation. This includes providing oversight throughout the process of inspecting and replacing the HS Shaft Aft seals, as detailed in section 6.7.10.1 of the technical manual. The key phases of work include:

3.1 Removal and Disassembly

The technician will guide the ship's force in the following steps:

- De-energizing and tagging out the main propulsion engines and turning gear motor.
- Securing diesel engines and the main reduction gear.
- Shifting the turning gear shift lock to the shaft lock position.
- Removing the clutch guard(s), clutch air manifold, clutch assembly, and adapter plate.

- Removing the clutch drum, quill shaft hub, and the hub from the pinion shaft extension.
- Removing oil supply and drain lines from the thru cap.
- Removing the high-speed thru cap and the bolts securing the seal cap.
- Removing the outer seal, spacer, and inner seal.

3.2 Cleaning

- Provide oversight for cleaning all disassembled parts with an appropriate solvent and soft brush, ensuring adherence to all safety warnings regarding flammable solvents.

3.3 Inspection

- Assist in the inspection of the removed components, specifically:
 - Seal Lands: Checking for scratches, breaks, or corrosion.
 - Rubber Seals: Checking for cracks, hard spots, or rips.

3.4 Repair

- Guide the replacement of any defective rubber seal components as required.

3.5 Installation

- Oversee the correct installation of the new components, including:
 - Installing the new inner seal, spacer, and outer seal, and securing the seal cap.
 - Installing the thru cap, oil supply/drain lines, and clutch guard adapter plate.
 - Installing the pinion shaft key and hub, followed by the clutch drum, clutch assembly, and air manifold.
 - Reinstalling the clutch guard.
 - Shifting the turning gear to the disengaged position and removing OUT OF SERVICE tags.
 - Verifying the lubrication system is properly reassembled with no leaks.

4.0 Contractor Responsibilities

- Provide one (1) qualified field service technician with expertise in the Philadelphia Gear 96DHCMG-CP reduction gear.
- The technician shall provide technical assistance, oversight, and quality assurance for the work described in Section 3.0.
- Submit root cause analysis and findings reports to ENS Marcos Saldarriaga. Contact information is found in the POC section of this document

5.0 Government Responsibilities

- The Coast Guard will provide all necessary materials, parts, seals, tools, and support personnel required to complete the work.
- The Coast Guard is responsible for providing a safe work environment.
- The Coast Guard is responsible for the final testing of the equipment upon completion of the work.

6.0 Place of Performance

- Onboard CGC WALNUT (WLB-205), 211 South Ave Ste C, Pensacola, FL 32508

7.0 Period of Performance

- The work is expected to take place from March 6, 2026 to March 16, 2026.

8.0 Points of Contact (POC)

- Primary
 - CWO3 Ryan Reed - 954-591-9397: Ryan.J.Reed@uscg.mil
- Secondary
 - ENS Marcos Saldarriaga - 667-419-5729: Marcos.A.Saldarriaga@uscg.mil